

Verification Results Summary

Outcomes of Test Protocols 1 and 2 — Home Audio System

Project	EE22005 Home Audio System (HAS) — Project Week 4
Group	Group HAS (BTA26, FS881, AC3847)
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Related protocols	PW4_HAS_VER_001 (Equalizer) and PW4_HAS_VER_002 (Class D)

Version	Date	Author	Description of changes
v1.0	01/05/2026	Group HAS (BTA26, FS881, AC3847)	Initial issue compiling results of Test Protocols 1 and 2, with explanations and conclusion.

1. Purpose

This document compiles the measured outcomes of the two formal verification test protocols defined for the Home Audio System: PW4_HAS_VER_001 covering the graphic equalizer and PW4_HAS_VER_002 covering the Class D amplifier. Each result is presented against its acceptance criterion, with a short explanation of any deviation. A combined conclusion follows in Section 5.

2. Test Protocol 1 — Equalizer Results

Verification was carried out on Day 2 of the project week using the function generator and four-channel oscilloscope. All measurements were captured as PNG screenshots and stored in the QMS Quality_Control folder per the naming convention [10].

Check item	Acceptance criterion	Measured	Pass / Fail	Notes
BPF1 (bass) centre frequency	100 Hz $\pm$ 15 %	108 Hz	Pass	Within +8 % of nominal — well

				inside $\pm 15\%$ spec.
BPF1 peak gain	12 dB ± 1 dB	11.6 dB	Pass	0.4 dB low; attributable to E12 rounding of feedback resistor.
BPF2 (mid) centre frequency	1 kHz $\pm 15\%$	0.94 kHz	Pass	-6% from nominal.
BPF2 peak gain	12 dB ± 1 dB	12.2 dB	Pass	Within ± 1 dB tolerance.
BPF3 (treble) centre frequency	8 kHz $\pm 15\%$	7.4 kHz	Pass	-7.5% ; likely op-amp GBW-limited at 8 kHz.
BPF3 peak gain	12 dB ± 1 dB	11.1 dB	Pass	Marginal pass; revisit if increased boost is required.
Full-EQ neutral ripple	< 4 dB	2.6 dB	Pass	Smooth summed response across audio band.
Full-EQ full-boost peaks	Three peaks $+12$ dB ± 1 dB	All three present	Pass	Three distinct peaks visible at expected centres.
Independent-control check	Adjacent change < 2 dB	1.4 dB max	Pass	Low cross-coupling between bands.

2.1 Explanation

All nine equalizer checks passed first time. The slight downward shift on BPF3 (-7.5% from 8 kHz) is consistent with the gain–bandwidth limitation of the TL072 at that gain setting, and remains comfortably inside the $\pm 15\%$ tolerance specified by S1. The two gain readings that fell below 12 dB nominal (11.6 dB and 11.1 dB) are within the ± 1 dB specification window; they reflect the deliberate decision to round design values to E12 standard components [5], an outcome already accepted in the choice matrix. Cross-coupling between bands was measured at a maximum of 1.4 dB at adjacent-band centres, well within the 2 dB threshold for usability specification S4.

3. Test Protocol 2 — Class D Results

Pre-power and powered verification was carried out on Day 3 (pre-power and gate-drive checks) and Day 4 (RLC filter and DC offset). The earphone hold-point was held closed until Day 5 morning.

Check item	Acceptance criterion	Measured	Pass / Fail	Notes
+V rail continuity	$< 1\ \Omega$	0.4 Ω	Pass	Stable; clean joints.
–V rail continuity	$< 1\ \Omega$	0.5 Ω	Pass	Within tolerance.

Analogue ground continuity	$< 1\ \Omega$	$0.3\ \Omega$	Pass	Solid ground return.
Half-bridge midpoint to RLC input	$< 1\ \Omega$	$0.6\ \Omega$	Pass	Acceptable.
+V rail to ground (insulation)	$> 1\ \text{M}\Omega$	$3.1\ \text{M}\Omega$	Pass	No short observed.
–V rail to ground (insulation)	$> 1\ \text{M}\Omega$	$2.7\ \text{M}\Omega$	Pass	No short observed.
Triangle wave frequency	$100\ \text{kHz} \pm 5\ \%$	$98.4\ \text{kHz}$	Pass	–1.6 % of nominal.
Triangle wave amplitude	Per design $\pm 10\ \%$	$5.8\ \text{V pp}$	Pass	Within tolerance.
PWM duty cycle modulation	Visible tracking	Tracking confirmed	Pass	Two-channel capture saved to Quality_Control.
Dead time — rising edge	$\geq 0.5\ \mu\text{s}$	$0.7\ \mu\text{s}$	Pass	Initially $0.2\ \mu\text{s}$ — failed.
Dead time — falling edge	$\geq 0.5\ \mu\text{s}$	$0.6\ \mu\text{s}$	Pass	After RC rework.
Shoot-through observed	None	None after rework	Pass	Initial RC value too small; reworked, then re-verified.
DC offset — no signal	$< 0.4\ \text{V}$	$0.12\ \text{V}$	Pass	Comfortable margin.
DC offset — low signal (100 mV)	$< 0.4\ \text{V}$	$0.18\ \text{V}$	Pass	Within margin.
DC offset — full signal	$< 0.4\ \text{V}$	$0.31\ \text{V}$	Pass	Closer to limit but within spec.

3.1 Explanation

The pre-power phase passed without any rework. The first attempt at the dead-time verification (Section 6.6 of Protocol 2) failed: rising-edge dead time measured at approximately $0.2\ \mu\text{s}$, with a brief overlap visible on the gate-drive capture. This matched the qualitative description of shoot-through risk in [3] and the rework instruction in the protocol. The dead-time RC values were increased per the schematic note (RC time constant raised from the original design value), the section was repeated in full, and dead time on both transitions was confirmed at $0.7\ \mu\text{s}$ and $0.6\ \mu\text{s}$ respectively — comfortably above the $0.5\ \mu\text{s}$ threshold and with no overlap observed. This single failure-and-rework cycle is recorded here as evidence that the verification protocol caught a real

defect that would otherwise have caused destructive shoot-through current under sustained operation.

The DC-offset hold-point was released only after all three signal conditions were measured below 0.4 V and signed off by the test lead (FS881). The full-signal case at 0.31 V is the tightest margin in the build and was re-checked after audio playback; it remained stable, supporting safe earphone connection per [1] and [9].

4. Compliance summary against target specifications

Spec	Description	Verification source	Outcome
S1	Equalizer frequency response (centres, gain, ripple)	Test Protocol 1, sections 6.1–6.5	Pass
S2	Class D switching efficiency and PWM correctness	Test Protocol 2, sections 6.4–6.6	Pass (after rework)
S3	RLC output filter ripple and offset	Test Protocol 2, section 6.7	Pass
S4	Independent per-band control	Test Protocol 1, section 6.6	Pass
S6	Hearing-safety output limits (DC offset)	Test Protocol 2, section 6.7 (hold-point)	Pass

5. Conclusion

All nine equalizer checks and all fifteen Class D checks ultimately passed against their acceptance criteria, satisfying target specifications S1, S2, S3, S4 and S6. The verification process caught one real defect — insufficient gate-drive dead time — and the documented rework path returned the system to compliance before the earphone-connection hold-point was released. The Home Audio System is therefore judged fit for full integration and final demonstration. Evidence files (oscilloscope screenshots, photographs and recorded measurements) are stored in the QMS Quality_Control folder under the file naming convention defined in [10].

References

- [1] International Electrotechnical Commission, Audio/video, information and communication technology equipment — Part 1: Safety requirements, IEC 62368-1:2023, 4th ed., May 2023.
- [3] D. Self, Audio Power Amplifier Design, 6th ed. Oxford, U.K.: Focal Press, 2013, ch. 12.
- [5] International Electrotechnical Commission, Preferred number series for resistors and capacitors, IEC 60063:2015.
- [9] World Health Organization, "Make Listening Safe," Geneva, Switzerland, 2015.
- [10] Department of Electronic and Electrical Engineering, EE22005 Engineering Practice and Design — Project Week 4 Brief, Univ. of Bath, 2026.